

Online Retail Sales Analytics

Project Overview

This project analyzes sales data of an online retail store using transactional data from 9994 purchases across various product categories and sub-categories. The goal is to uncover insights into sales based on categories/sub-categories, monthly sales trend, unprofitable sub-categories to guide strategic business decisions.

Dataset Summary

- Rows: 9994
- columns: 21
- Customer Demographics (Customer ID, Customer Name, Country, City, State, Postal Code, Region)
- Purchase details (Order ID, Order Date, Ship Date, Ship Mode, Product ID, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit)
- Missing Data: 0

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

- **Missing Data Handling and Duplication Check:** Checked for null values and duplication, but no null values or duplicated values found.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Create derived columns:** New columns ('shipping_days', 'profit_margin', 'order_year', 'order_month') were created for answering Business Questions.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

Data Analysis using SQL

1. **Total sales and profit by region, category, sub-category** – compared the total sales and profits based on region of customers and category and sub category of the product.

region text	category text	sub_category text	total_sales numeric	total_profit numeric
West	Technology	Copiers	49749.24	19327.24
East	Technology	Copiers	53219.46	17022.84
West	Technology	Accessories	61114.12	16484.60
West	Office Suppli...	Binders	55961.11	16096.80
Central	Technology	Copiers	37259.57	15608.84
Central	Technology	Phones	72403.28	12323.03
East	Technology	Phones	100614.98	12314.69
West	Office Suppli...	Paper	26663.72	12119.24
East	Office Suppli...	Binders	53498.00	11267.93
East	Technology	Accessories	45033.37	11195.86
South	Technology	Phones	58304.44	10767.28
East	Furniture	Chairs	96260.68	9357.77
West	Technology	Phones	98684.35	9110.74
East	Office Suppli...	Paper	20172.60	9015.37
West	Office Suppli...	Storage	70532.85	8645.32

2. **Monthly sales trends** – Sales and profits on each month is compared.

month timestamp without time zone	monthly_sales numeric	monthly_profit numeric
2014-01-01 00:00:00	14236.90	2450.19
2014-02-01 00:00:00	4519.89	862.31
2014-03-01 00:00:00	55691.01	498.73
2014-04-01 00:00:00	28295.35	3488.84
2014-05-01 00:00:00	23648.20	2738.71
2014-06-01 00:00:00	34595.13	4976.52
2014-07-01 00:00:00	33946.39	-841.48
2014-08-01 00:00:00	27909.47	5318.11
2014-09-01 00:00:00	81777.35	8328.10
2014-10-01 00:00:00	31453.39	3448.26
2014-11-01 00:00:00	78628.72	9292.13
2014-12-01 00:00:00	69545.62	8983.57
2015-01-01 00:00:00	18174.08	-3281.01
2015-02-01 00:00:00	11951.41	2813.85
2015-03-01 00:00:00	38726.25	9732.10
2015-04-01 00:00:00	34408.93	4407.50

3. **Consistently unprofitable sub-categories-** Most unprofitable products based on sub-category is identified.

sub_category text	total_profit numeric
Tables	-17725.48
Bookcases	-3472.56
Supplies	-1189.10

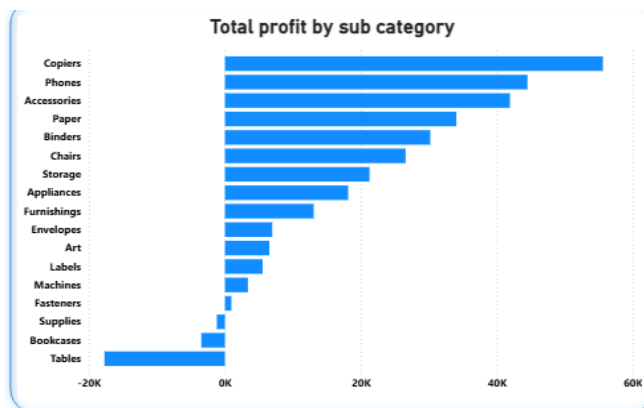
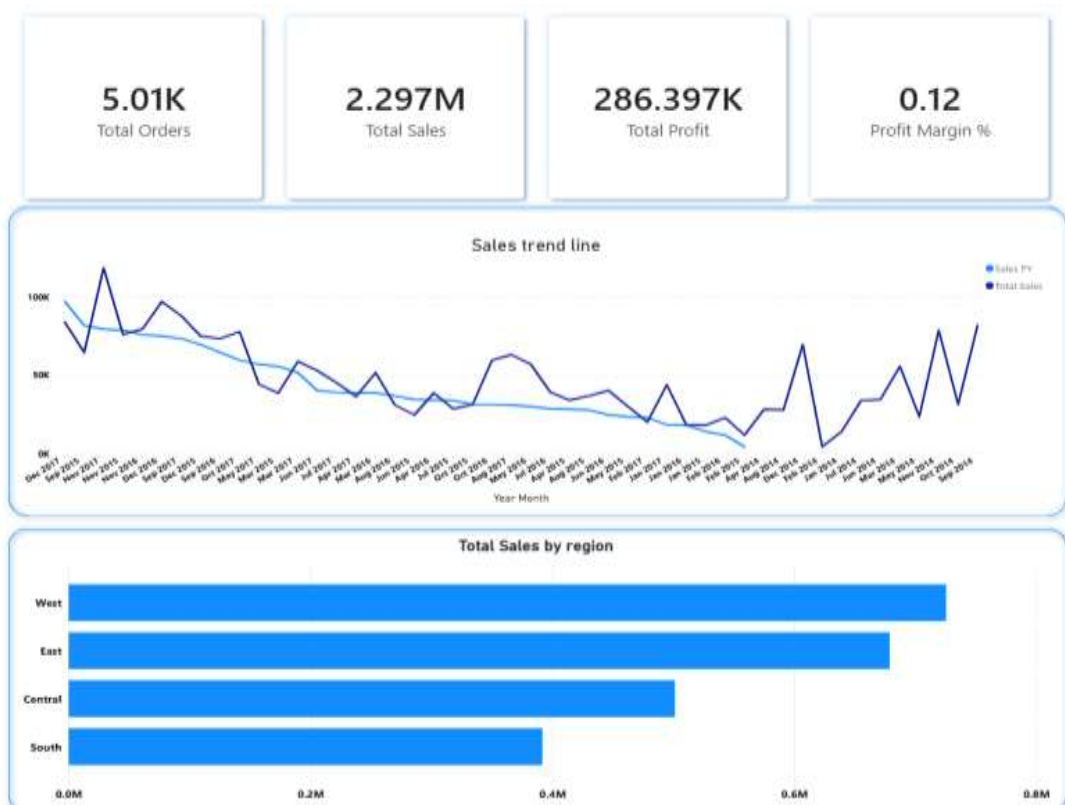
4. **High-sales but negative-profit orders** – Orders that has high sales value but that has negative profits are identified.

order_id text	product_name text	sales double precision	discount double precision	profit double precision
CA-2014-1453...	Cisco TelePresence System EX90 Videoconferencing Unit	22638.48	0.5	-1811
CA-2014-1368...	Lexmark MX611dhe Monochrome Laser Printer	8159.952	0.4	-135
US-2017-1681...	Cubify CubeX 3D Printer Triple Head Print	7999.99	0.5	-3839
CA-2017-1583...	Martin Yale Chadless Opener Electric Letter Opener	4663.736	0.2	-1049
CA-2016-1081...	Cubify CubeX 3D Printer Double Head Print	4499.985	0.7	-659
CA-2017-1188...	HON 5400 Series Task Chairs for Big and Tall	4416.174	0.3	-63
CA-2015-1166...	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	4297.644	0.4	-1862
CA-2014-1282...	DMI Eclipse Executive Suite Bookcases	4007.84	0.2	-9
CA-2016-1039...	High Speed Automatic Electric Letter Opener	3930.072	0.2	-786
US-2015-1506...	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	3083.43	0.5	-1665
US-2014-1069...	Lexmark MX611dhe Monochrome Laser Printer	3059.982	0.4	-50
CA-2014-1272...	HP DesignJet T520 Inkjet Large Format Printer - 24" Color	2624.985	0.5	-944
CA-2017-1348...	Lexmark MX611dhe Monochrome Laser Printer	2549.985	0.7	-33
US-2014-1069...	Xerox WorkCentre 6505DN Laser Multifunction Printer	2519.958	0.4	-251
CA-2015-1397...	HON 5400 Series Task Chairs for Big and Tall	2453.43	0.3	-9

5. **Repeat vs one-time buyers-** The number of first time buyers and repeat customers are identified.

customer_type text	num_customers bigint
Repeat buyer	781
One-time buyer	12

DashBoard in Power BI



Loss Leader orders table

order_id	product_name	sales	Sum of discount	Sum of profit
CA-2016-108196	Cubify CubeX 3D Printer Double Head Print	4,499.99	0.70	-6,599.98
US-2017-168116	Cubify CubeX 3D Printer Triple Head Print	7,999.98	0.50	-3,839.99
CA-2014-169019	GBC DocuBind P400 Electric Binding System	2,177.58	0.80	-3,701.89
CA-2017-134845	Lexmark MX611dhe Monochrome Laser Printer	2,549.99	0.70	-3,399.98
US-2017-122714	Ibico EPK-21 Electric Binding System	1,889.99	0.80	-2,929.48
CA-2015-147830	Cubify CubeX 3D Printer Double Head Print	1,799.99	0.70	-2,639.99
CA-2017-131254	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	1,525.19	0.80	-2,287.78
CA-2015-116638	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	4,297.64	0.40	-1,862.31
CA-2016-130946	GBC DocuBind P400 Electric Binding System	1,088.79	0.80	-1,850.95
CA-2014-145317	Cisco TelePresence System EX90 Videoconferencing Unit	22,638.48	0.50	-1,811.08
US-2015-150630	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	3,083.43	0.50	-1,665.05
CA-2014-165309	GBC DocuBind TL300 Electric Binding System	896.99	0.80	-1,480.03
CA-2014-139892	Lexmark MX611dhe Monochrome Laser Printer	8,159.95	0.40	-1,359.99
US-2017-120390	GBC DocuBind P400 Electric Binding System	1,633.19	0.70	-1,306.55
CA-2017-128363	GBC DocuBind TL300 Electric Binding System	1,614.58	0.70	-1,237.85
CA-2015-140025	Hoover Upright Vacuum With Dirt Cup	463.25	0.80	-1,181.28
Total			899.74	-1,56,131.29